Strategies and best practices for organising and formatting data

Liz Smy, Beate Lichtwardt, Sharon Bolton, and Cristina Magder

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Introduction

High quality data are well organised, structured, named and versioned and the authenticity of master files identified. Using standard and interchangeable or open lossless data formats ensures long-term usability of data.

We are going to look at a number of topics around formatting your data and how planning and documenting these processes, and being aware of the best practices and common areas to check, will help you manage your data throughout the data lifecycle.

We will look at:

- **Version control and authenticity** (best practice and methods)
- **File formats and data conversion** (what to consider when choosing a file format, possible loss or change of information during conversion, and some common checks)
- **Variable formatting** (checking variable types and variable formats, and specifically issues with decimal places)
- **Organising files and folders** (best practice)
- **Quality assurance** (data checking and documentation)

It is worth mentioning these topics are applicable to all data types, not just longitudinal data. However, they are particularly important to longitudinal data to ensure good practice is replicated with each new wave of data.

1. Version control and authenticity

Data management processes will inevitably result in a number of edits to the data. In further sessions we will be looking into changes which might be made in terms of documenting your data, such as labelling or creating derived variables, as well as issues around anonymisation. All these edits are likely to result in different **versions** of the data being created. Different versions are useful as they allow you to step back to an earlier version of the data before specific changes were made. It is important to preserve the **authenticity** of the original data which is why planning a **version control strategy** is essential.

It is important to ensure that different versions of files, files held in different formats or locations, and information that is cross-referenced between files are all subject to version control.

1.1 Version control strategy

Version control planning will help you to clearly distinguish between different versions of the data and keep track of the changes and modifications. It will prevent unauthorised amendments of files and potential loss of information, and in so doing preserve the data authenticity.

A suitable version control strategy depends on whether files are used by single or multiple users, in one or multiple locations, and whether versions across users or locations need to be synchronised or not. Synchronising files due to the file being saved in multiple locations or being used by multiple users, adds an extra complexity but this can be planned for.

1.2 Best practice for version control

It is best practice to implement the following actions:

- Decide how many versions of a file to keep, which versions to keep, for how long and how to organise versions.
- Identify milestone versions to keep, e.g. major versions rather than minor versions (keep version 02-00 but not 02-01).
- Uniquely identify different versions of files using a systematic naming convention, such as using version numbers or dates.
- Record changes made to a file when a new version is created.
- Record relationships between items where needed, for example, between code and the data file it is run against; between data file and related documentation or metadata; or between multiple files.
- Track the location of files if they are stored in a variety of locations.
- Regularly synchronise files in different locations, by using dedicated software such as MS SyncToy.
- Identify a single location for the storage of milestone and master versions.

Once a version control strategy is in place, it can be replicated with each new wave of data, so is especially important to longitudinal data.

1.3 Methods for version control

Version control can be done through the following:

- The date recorded in the file name or within the file, for example, HealthTest-2008-04-06.
- Version numbering in the file name, for example, HealthTest-00-02 or HealthTest_v2.
- A file history, version control table or notes included within a file, where versions, dates, authors and details of changes to the file are recorded.
- Version control facilities within the software used.
- Using versioning software, e.g. Subversion.
- Using file-sharing services, such as Dropbox or Google Docs.
- Controlling rights to file-editing.

- Manual merging of entries or edits by multiple users.

1.4 Example version control (tables)

The following examples show how version control can be done.

The first table shows how information can be recorded about a document.

The second table illustrates how information about each version of the document can be recorded.

Document Control:

Title:	**COVID-19 PCR tests in Essex schools**
File:	PCRResultsEssex_00_05
Description:	Results data of 180 COVID-19 PCR tests carried out in 2 schools in January 2021
Created By:	Chris Wilkins
Maintained By:	Sally Water
Created:	11/01/2023
Last Modified:	30/03/2023
Based on:	PCRDatabaseDesignEssex_03_00

Version History:

Version	Responsible	Notes	Last amended
00_05	Vani Yussun	Entries checked by VY, independent from SK	30/03/2023
00_04	Steve Knights	Entries checked by SK	12/03/2023
00_03	Karin Millys	Test results 121-180 entered	25/02/2023
00_02	Karin Millys	Test results 61-120 entered	05/02/2023

1.5 Data authenticity

It is very easy to copy or alter digital information, so it is important to be able to demonstrate the authenticity of data and prevent unauthorised access to data that may potentially lead to inadvertent changes.

Best practice to ensure authenticity is to:

- Keep a single master file of data.
- Assign responsibility for master files to a single project team member.
- Regulate write access to master versions of data files.
- Record all changes to master files.
- Maintain old master files in case later ones contain errors.
- Archive copies of master files at regular intervals.
- Develop a formal procedure for the destruction of master files.

2. File formats and data conversion

The choice of file format is important for the reusability of the data and long-term preservation. The choice of file format is likely to vary at different stages during the research lifecycle, so early planning can be useful.

The choice of file formats is related to the '**R**eusability' **FAIR** Principle, specifically the R1.3. (Meta)data meet domain-relevant community standards.

Findable
Accessible
Interoperable
Reusable

This principle recommends following community standards or best practices for both data and metadata, to try to increase the usability, interoperability and long-term preservation of data. It is important to consider this when choosing a suitable file format, and if applicable, a metadata system and schema.

2.1 Considerations when choosing a file format

In practice, there are a number of questions to help you consider which file format is suitable. We will now look at each of these questions in more detail.

- **What format is best suited for data creation?**

The format in which you receive the data from the survey team might be dependent on the hardware used, the availability of software, or be determined by discipline-specific standards and customs.

- **What format is best suited for data analyses and other planned uses?**

The format choice is likely to be related to the specific software that you intend to use for data analysis. This might follow discipline-specific standards and customs, or be linked to the software skills or preference of the research team.

- **What format is best suited for long-term sustainability and sharing of data?**

For many formats, there is a danger that they will become obsolete in the future, which would make the data impossible to read and interpret. Despite the backward compatibility of many software packages to import data created in previous software versions and the interoperability between competing popular software programmes, the safest option to guarantee long-term data access is to convert data to standard or open formats.

- **Should you choose an open versus a proprietary format?**

When selecting file formats for archiving and sharing, you should consider whether to use a proprietary file format, or a more 'open' and interoperable one. This may depend on the requirements of your collaborators, or the repository that will archive your data. If you plan to make data publicly available, you should choose an open, platform-independent format where possible.

Proprietary file formats are those owned by a company that claims intellectual property rights for the use of the software by granting licenses. Their specifications are generally not publicly available and there is a risk that future developments and software updates could impact the usability of files created in previous versions. While the most widely-used proprietary formats will have better long-term sustainability, obsolescence or backward incompatibility may still be an issue. Well-known standard proprietary formats include Microsoft Office software products, (e.g. MS Word and MS Excel), or the popular SPSS or Stata statistical packages. Proprietary formats are often not readable in other software packages unless import facilities exist, and changes may occur during conversion. For example, Stata format data files can be imported into SPSS, but the two software packages have different specifications for missing values, file and variable metadata and other elements, which may result in information loss.

International open standards have been developed over time and are standardized across organizations to ensure a degree of interoperability between platforms. The files produced to these standards are known as open format files. Examples of these include:

- The American Standard Code for Information Interchange character encoding standard (ASCII), to which CSV (comma-separated values), TSV (tab-delimited format) and fixed-with text data files usually conform.
- Tagged Image File Format (TIFF, or TIF) image files.
- PDF/A, an ISO-standardized version of the Portable Document Format (.pdf), for use in the archiving and long-term preservation of electronic documents.

The specifications for open formats are publicly available and are widely recognised across countries and organisations. Other open standards include OpenDocument Format (ODF) and XML. These formats may still require software that is capable of reading them, such as a PDF reader, but free, open-source packages for these formats are generally widely available.

- **Should the format be lossy or not?**

Some file formats are created specifically to save space, but result in a reduction of information and data quality. When this occurs, we can describe file formats as 'lossy', as opposed to 'lossless' (where no information is lost). Lossy formats save space by removing detailed information assumed (rightly or wrongly) to be unimportant. For example, the lossy format JPEG removes fine detail in images, whilst the lossless format TIFF keeps everything. Also, repeatedly editing and saving files in lossy format results in a greater loss of information, so this should be taken into consideration.

- **Is the format suitable for conversion?**

There may be valid reasons why the data cannot be converted to other formats. Even when commonly used proprietary formats are used, it is good practice to create an open format file in addition to ensure long-term preservation, even though this may result in a less 'user-friendly' format. For example, the creation of a CSV version of each SPSS or Stata format file is a good policy, although such files will not contain label metadata, which will need to be saved separately.

As we mentioned earlier, the answers to these questions might result in different file formats being suitable for different stages during the research lifecycle. Different versions should of course be documented and included in the version control strategy.

2.2 Data conversion and possible loss or change of information

Any changes to the data should always be checked due to the risk of error or information loss. It is best practice for data conversion to be completed by someone who is familiar with the data and can check for any unwanted changes.

Some of the common issues are:

Quantitative data	Qualitative data
Loss or change of metadata	Loss of formatting
Change of variable formats	Loss of headers/footers
Change in missing value definitions	
Loss of macros or formulae	
Differences in character coding	
Data truncation	

Some of the common areas which can change in **quantitative data** are:

Loss or change of metadata: Variable or value labels may not be transferred, for example from SPSS to CSV, or they might be truncated due to different allowed character lengths between software packages.

Change of variable formats: Several variable characteristics can be changed: the variable type (i.e. whether the variable is treated as numeric, date, or string format), variable width (including the number of decimal places), or variable measurement level (i.e. whether the variable is treated as scale, nominal, or ordinal) can be altered.

Change in missing value definitions: The way missing values are defined or set can change. Values might be set to missing in one data file format but not in another. Different software packages treat missing values differently.

Loss of macros or formulae: Software specific information might be lost in conversion. For example, macros or formulae included in an Excel spreadsheet would be lost when converting to CSV format. Other ways of recording or preserving this information should be considered.

Differences in character coding: Software packages may use different character encoding schemes which means some characters are not supported. This can cause problems, as characters in variable and value labels may not transfer correctly.

Data truncation: Occasionally truncation of the data can occur. This may be due to software incompatibility or error.

Some of the common areas which may change in **qualitative data** are:

Loss of formatting: Editing features, such as highlighting, bold or italic text might be lost. This can reduce the usability of the data.

Loss of headers/footers: Header and footer sections in documents are often used to identify interview transcripts but are not supported by all text writer software.

2.3 Data conversion methods

We are not going to look at this in detail, but the method of data conversion will depend on your original data format and the target data format. Most software packages have an 'export' or 'save as' function which will allow the data to be saved in a new format. You may also be able to import either an open or proprietary format file into your chosen software package. For example, SPSS can import CSV format files and provides a step-by-step conversion process where you can tell SPSS how to treat certain aspects of the data, for example whether - variable names are included on the first row of the data file. At the UK Data Archive, we also use the proprietary software package Stat/Transfer, which is a fast, reliable, and convenient data transfer tool. Your planned data repository is likely to be able to help you with data conversion if you are unsure.

3. Variable formatting

We touched on variable formatting briefly when discussing awareness about potential changes and losses to the data during file format transfers, but it is useful to consider this topic in more detail. We frequently find incorrect variable formatting which suggests variable details may not be checked before deposit. Incorrect variable formatting can lead to incorrect data use and sometimes even reduces the usability of the data.

3.1 Variable types

The type of variables included in your data will be influenced by the method of data collection, the software used and the subject of the variable. We are not going to look at all variable types as different software packages deal with variable types in different ways.

However, a very simple but important check which should be made is to determine whether the data are to be treated as **string** or **numeric**. If a variable includes any letters or special characters it will be set as a string variable in most statistical packages, which can dramatically reduce the usability of the data, as numeric functions will not be available. Use of non-numeric characters should be avoided where possible and only used in dedicated string variables or variable metadata (the variable and value labels). String variables are useful for including 'other' responses and can allow for rich open-ended question responses to be shared, but they should be checked and anonymised as appropriate.

If the data have been transferred from a different format (such as CSV to SPSS), it is important to check the numeric variables are still in this format. Sometimes the text value labels are transferred instead of the numerical values, which would change the variable from numeric to string format and impact on how the variable could be analysed.

3.1.1 Measurement levels (numeric variables)

Numeric variables also need to be checked to ensure the measurement level is correctly defined. Different software packages deal with measurement levels differently but it is something you should be aware of because it is vital for how the variable can be used for analysis.

We are going to look at the different measurement levels used by SPSS and then Stata.

SPSS has three defined measurement levels: Nominal, Ordinal, and Scale.

3.1.1 Numeric variables – measurement levels (SPSS)

Nominal:

Categorical data which have no order, and one value is not greater than another.

sex Sex

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	1 Male	8827	45.5	45.5	45.5
	2 Female	10584	54.5	54.5	100.0
	Total	19411	100.0	100.0	

Citation:
University of Manchester. Cathie Marsh Centre for Census and Survey Research. ESDS Government. (2003). *British Crime Survey, 2000: Teaching Dataset.* [data collection]. UK Data Service. SN: 4740, DOI: http://doi.org/10.5255/UKDA-SN-4740-1

- **Nominal:** Categorical data which has no order and one value is not greater than another. The actual numerical value is not meaningful. For example, 1=Male and 2=Female.

3.1.1 Numeric variables – measurement levels (SPSS)

Ordinal: Categorical data which can be ordered and ranked. Ordinal measures are common in survey data due to the use of closed questions and rating scales.

walkdark Feel safe while walking alone after dark

	Frequency	Percent	Valid Percent	Cumu Per
1 Very safe	4737	24.4	24.5	
2 Fairly safe	8172	42.1	42.3	
3 A bit unsafe	4030	20.8	20.9	
4 Very unsafe	2380	12.3	12.3	
Total	19319	99.5	100.0	
8 refusal	1	.0		
9 don't know	91	.5		
Total	92	.5		
	19411	100.0		

Citation:
University of Manchester. Cathie Marsh Centre for Census and Survey Research. ESDS Government. (2003). *British Crime Survey, 2000: Teaching Dataset.* [data collection]. UK Data Service. SN: 4740, DOI: http://doi.org/10.5255/UKDA-SN-4740-1

- **Ordinal**: Categorical data which can be ordered and ranked. Ordinal measures are common in survey data due to the use of closed questions and rating scales. For

example, the question 'How safe do you feel walking alone in this area after dark?' has four response options: 1.Very Safe, 2. Fairly Safe, 3. A bit unsafe, and 4. Very unsafe.

3.1.1 Numeric variables – measurement levels (SPSS)

Scale: Numerical data that indicate differences in magnitude. Common examples include age and income. The distance between levels ('intervals') is equivalent across the range of values.

AGE Age of respondent

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	16	648	1.9	1.9	1.9
	17	566	1.6	1.6	3.5
	18	507	1.5	1.5	4.9
	19	417	1.2	1.2	6.1
	20	376	1.1	1.1	7.2
	21	380	1.1	1.1	8.3
	22	413	1.2	1.2	9.5
	23	421	1.2	1.2	10.7
	24	450	1.3	1.3	12.0
	25	424	1.2	1.2	13.2
	26	450	1.3	1.3	14.5
	27	436	1.2	1.2	15.7
	28	514	1.5	1.5	17.2
	29	564	1.6	1.6	18.8
	30	586	1.7	1.7	20.5
	31	630	1.8	1.8	22.3
	32	652	1.9	1.9	24.1
	33	628	1.8	1.8	25.9
	34	640	1.8	1.8	27.8

Citation:
University of Manchester, Cathie Marsh Institute for Social Research (CMIST), UK Data Service. (2023). *Quarterly Labour Force Survey, January-March 2023: Teaching Dataset.* [data collection]. Office for National Statistics (ONS). [original data producer(s)]. Office for National Statistics (ONS). SN: 9172. DOI: http://doi.org/10.5255/UKDA-SN-9172-1

- **Scale:** Numerical data that indicate differences in magnitude. Common examples include age and income. The distance between levels ('intervals') is equivalent across the range of values. For example, a 50 year old is ten years older than a 40 year old and a 40 year old is ten years older than a 30 year old.

STATA has a slightly different approach and has two common categories: Categorical and Continuous.

- **Categorical:** Stata does not distinguish between nominal and ordinal data and simply describes all data with defined groups as categorical. So the examples of 1=Male and 2=Female and 1.Very Safe, 2. Fairly Safe, 3. A bit unsafe, and 4. Very unsafe would both be defined as categorical in Stata.

- **Continuous:** This is equivalent to a Scale variable in SPSS. Stata refers to the variable measuring something such as e a person's age, height, or weight; a city's population or land area; or a company's revenues or costs.

3.2 Variable formats

Once you have checked the variable type is correct, it is important to check the format of the variable is correct too. This will ensure the variable presents the data in the most suitable way and will ensure the correct statistical analysis can be performed. The software used will influence the variable formats that need to be checked, so we will not look at this in further detail. However we do want to mention the importance of checking decimal places.

It is good practice to check decimal places are correct for each variable. Although this does not change your actual data values, the variable formats determine how your data are displayed. A common issue we see is with variables including too many decimal places.

In this example, we can see the format of the age variable is set to 2 decimal places. This is incorrect as age has been recorded as a whole number. The decimal places are redundant and can cause problems for future use of the data.

	caseno	Age	Gender
1	2353	17.00	Female
2	2375	19.00	Male
3	2397	20.00	Female
4	2067	20.00	Male
5	2309	17.00	Male
6	1341	19.00	Male
7	1583	18.00	Female
8	1825	19.00	Female
9	2573	20.00	Male
10	2749	20.00	Female
11	2815	17.00	Female
12	1495	19.00	Male
13	1539	18.00	Female
14	1605	17.00	Male

Data View | Variable View

It is important to check decimal places are transferred correctly when converting data to a new file format.

3.2.1 Variable formats - extra longitudinal considerations

Longitudinal data is more likely to be linked or matched to other datasets, so it is important to ensure variables are consistent between waves and all datasets.

It is good practice to ensure variables which are repeated across waves or datasets have the same formats. This helps both internal and external users of the data and reduces the amount of formatting users will need to complete to use the data.

- Variable types should be the same, i.e. numeric or string.
- Variable formats should be the same i.e. widths, decimal places, measurement levels.

If you look at the examples shown you will see that the variable caseno, which is a key ID variable used for matching, has different variable formats in the two examples. The top variable is in numeric format and the second example is in string format. You would not be able to successfully match the two datasets using caseno, as the key matching variable, due to the different formats. You would have to amend the format of one variable first, so they match.

Similarly, the age variable is set to two decimal places in the top example and has no decimal places in the bottom example. While this variable might not be used for data matching, it is good practice to check the decimal places for sense and consistency. Age is likely to be reported as a single year so including two decimal places is meaningless if they are just zeros. It is helpful to users if variables with the same name have the same variable formats in all datasets.

	Name	Type	Width	Decimals	Label	Values	Missing	Columns	Align	Measure
1	caseno	Numeric	8	0	ID	None	None	8	Right	Nominal
2	Age	Numeric	8	2	Age of respondent	None	None	8	Right	Ordinal
3	Gender	Numeric	8	0	Gender of respondent	{0, Male}...	None	8	Right	Nominal

	Name	Type	Width	Decimals	Label	Values	Missing	Columns	Align	Measure
1	caseno	String	8	0	ID	None	None	8	Left	Nominal
2	Age	Numeric	8	0	Age of respondent	None	None	8	Right	Ordinal
3	Gender	Numeric	8	0	Gender of respondent	{0, Male}...	None	8	Right	Nominal

Activity: Recap quiz on variable measurement levels

Q: How should each of the following variables be defined in SPSS and Stata?

Age, ethnicity, annual income (banded), life satisfaction, height (cm), marital status, age (banded), general health, and monthly income (£).

Recap quiz on variable measurement levels

Variable	Stata	SPSS
Age (single year)		
Ethnicity		
Annual income (banded)		
Life satisfaction		
Height (cm)		
Marital status		
Age (banded)		
General health		
Monthly income (£)		

The options are:
STATA: Categorical, Continuous
SPSS: Nominal, Ordinal, Scale

Recap quiz - answers

Variable	Stata	SPSS
Age (single year)	Continuous	Scale
Ethnicity	Categorical	Nominal
Annual income (banded)	Categorical	Ordinal
Life satisfaction	Categorical	Ordinal
Height (cm)	Continuous	Scale
Marital status	Categorical	Nominal
Age (banded)	Categorical	Ordinal
General health	Categorical	Ordinal
Monthly income (£)	Continuous	Scale

The options are:
STATA: Categorical, Continuous
SPSS: Nominal, Ordinal, Scale

4. Organising files and folders

Sensible file names and well-organised folder structures make it easier to find and keep track of data files. This is particularly important when collaborating with others.

4.1 File names

A file name is a principal identifier for a file. Good file names provide useful clues to the content, status and version of a file. They should uniquely identify a file and help in classifying and sorting files. File names that reflect the file content also facilitate searching and discovering files. In collaborative research, it is essential to keep track of changes and edits to files via the file name. File names should be independent of the location of the file on a computer. Whilst computers add basic information and properties to a file, such as file type, date and time of creation and modification, this is not reliable data management. It is better to record such essential information in the file name.

Software can be used for bulk renaming of files and folders in a consistent way. This can be helpful when multiple files have been generated automatically, for example images from a digital camera, or when you wanted to apply a change to all file names such as replacement of spaces with an underscore (_). There are various tools available but at the UK Data Archive we use a powerful open-source package, Bulk Rename Utility.

Common elements that should be considered when developing a file naming strategy are:

- version number
- date of creation
- name of creator
- description of content
- publication date
- project number.

Best practice for file name creation is to:

- Create meaningful but brief names.
- Use file names to classify types of files.
- Use dates in the format YYYY-MM-DD.
- Avoid capitalisation or 'camel case' where possible, as some computer platforms may be case-sensitive (e.g. Unix).
- Avoid using spaces, dots and special characters (& or ? or !).
- Use hyphens (-) or underscores (_) to separate elements in a file name.
- Avoid very long file names.
- Reserve the 3-letter file extension for application-specific codes of file format (e.g. .docx, .xlsx, .mov, .tif).
- Include versioning within file names where appropriate.
- Review file names for archived versions to ensure they do not contain any confusing, irrelevant information (e.g. versioning, misleading description).

For longitudinal data, the approach should be replicated across waves. This helps both internal and external users as they will gain familiarity with the file naming convention.

Examples of **poor file naming**:

File name	Problem
project survey data.dta	The file name does not include any descriptive information, such survey or project name, dates or version, and includes spaces.
Health&Safety Procedures1.	The file name includes capitalisation, spaces and special characters and has no file extension.
First interview with sally on 1st Feb 23.doc	The file name is long and includes too much information in text format. The date should be in the format YYYY-MM-DD and it would be beneficial to use numbers or identifiers instead of text and a name.

Examples of **useful file names**:

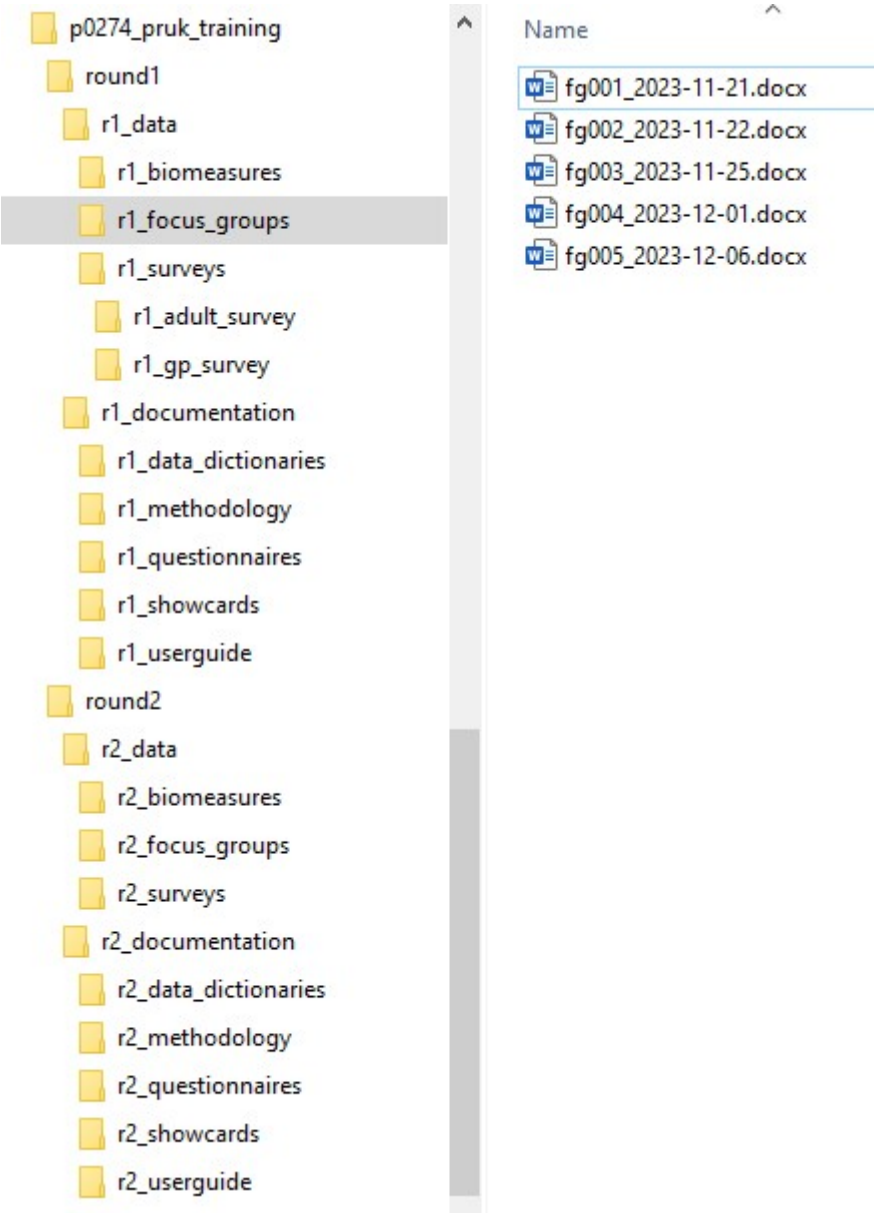
File name	Meaning
ess_wave8_survey_e01v02.sav	Dataset from the European Social Survey wave 8, first edition, second version.
fg1_cons_2022-02-12.rtf	Transcript of the first focus group with consumers, that took place on 12 February 2022.
int024_ap_2023-06-05.doc	Interview with participant 024, interviewed by Anne Parsons on 5 June 2023.
shes_21_dataset_documentation_v2.pdf	Scottish Health Survey 2021 dataset documentation second version.

4.2 Folder structure

It is important to think carefully how best to structure files in folders, in order to make it easy to locate and organise files and versions. When working in collaboration with others, the need for an orderly structure is even greater. The hierarchy for files will be project-specific but it is always useful to hold data and accompanying documentation in separate folders.

Example folder structure

This example is a detailed, 'deep' folder structure but shows how different elements of the project can be organised. The data and documentation files are held in separate high-level folders. Data files are further organised according to data type and then according to research activity. Documentation files are organised also according to type of documentation file and research activity.



5. Quality assurance

Quality assurance is a reoccurring theme throughout this training, but as noted above, it is very important to check and document any changes made to your data, which will provide a history, version control and provenance trail to help quality assure your data.

5.1 Data checking

It is more than likely your data have been checked a number of times before you receive the -files, but you might still have a role in validating the data. If you plan to share the data, it is important that a number of checks are completed.

This may include:

- Double-checking coding of observations or responses and out-of-range values.
- Checking data completeness.
- Adding variable and value labels where appropriate.
- Double entry of data.
- Statistical analyses, such as frequencies, means, ranges or clustering to detect errors and anomalous values.
- For qualitative data, correcting errors made during transcription.

5.2 Documentation

As noted several times already, any changes made to the data should be documented. We will also cover this in the Session on 'Effective methods and practices for documenting data', but we just want to remind you of the importance of documenting change in relation to formatting your data.

Key areas to document are:

- **The methods or process used to transfer between different file formats.** It is good practice to save the syntax or code so the change can be replicated. If syntax is not used or appropriate, then the process should still be recorded so it can be replicated.
- **Data differences identified between file formats.** There is likely to be some differences in the data in different file formats. These could be differences in how missing values are treated or differences in the value labelling. These differences need to be recorded to ensure everyone is aware of any differences.
- **Any changes made to the data.** It is good practice to document any changes made to the data, however small or irrelevant they might seem. It is important to ensure the changes can be reproduced. This could be useful for future editions and also to ensure the same steps are completed for future waves.

This brings us back to the start of this session, where we discussed how version control planning can help distinguish between different versions of the data and keep track of the inevitable changes and modifications required to your data. It is important to be able to demonstrate the authenticity of your data, and documenting changes plays a key role in this process.

6. Resources

UKDS Learning Hub Research data management
ukdataservice.ac.uk/learning-hub/research-data-management/

CESSDA Data Management Expert Guide (DMEG)
dmeg.cessda.eu/

Managing and sharing data: best practice for researchers
dam.ukdataservice.ac.uk/media/622417/managingsharing.pdf

Supporting FAIR data: File formats (Data Archiving and Networked Services (DANS-KNAW))
zenodo.org/records/10571099

Buckley, Jen and Sarah King-Hele (2015). Using survey data. UK Data Service, University of Essex and University of Manchester.
ukdataservice.ac.uk/app/uploads/using-survey-data.pdf

Stata Manual - 26 Working with categorical data and factor variables
www.stata.com/manuals/u26.pdf

ukdataservice.ac.uk

pruk.ac.uk

datasharing@ukdataservice.ac.uk

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